

AFRILANG Stdlib

std/c/graphbfs

Bibliothèque AFRILANG ``std/c/graphbfs`` (niveau complex, catégorie complex). Elle expose 7 fonction(s) :
bfsDistances, bf

```
`import "std/c/graphbfs" . `use graphbfs`
```

- ``bfsDistances(adj list number, n number, start number) ? list number``
- ``bfsOrder(adj list number, n number, start number) ? list number``
- ``bfsReachable(adj list number, n number, start number) ? number``
- ``bfsLayers(adj list number, n number, start number) ? list number``
- ``isConnected(adj list number, n number) ? bool``
- ``shortestPathLen(adj list number, n number, start number, goal number) ? number``
- ``multiSourceBfs(adj list number, n number, sources list number) ? list number``

Category: complex | Tier: complex | Functions: 7

FRANCAIS

1. Vue d'ensemble

Bibliothèque AFRILANG ``std/c/graphbfs`` (niveau complex, catégorie complex). Elle expose 7 fonction(s) : `bfsDistances`, `bf`

```
`import "std/c/graphbfs"` · `use graphbfs`  
- `bfsDistances(adj list number, n number, start number) ? list number`  
- `bfsOrder(adj list number, n number, start number) ? list number`  
- `bfsReachable(adj list number, n number, start number) ? number`  
- `bfsLayers(adj list number, n number, start number) ? list number`  
- `isConnected(adj list number, n number) ? bool`  
- `shortestPathLen(adj list number, n number, start number, goal number) ? number`  
- `multiSourceBfs(adj list number, n number, sources list number) ? list number`
```

2. Installation & import

Ajoutez le module à votre programme :

```
import "std/c/graphbfs"  
use graphbfs
```

Niveau : complex · Catégorie : Modules complex (c/)
Domaines avancés ? graphes, NLP, réseaux complexes.

3. Référence API

```
? bfsDistances(adj list number, n number, start number) ? list  
? bfsOrder(adj list number, n number, start number) ? list  
? bfsReachable(adj list number, n number, start number) ? number  
? bfsLayers(adj list number, n number, start number) ? list  
? isConnected(adj list number, n number) ? bool  
? shortestPathLen(adj list number, n number, start number, goal number) ? number  
? multiSourceBfs(adj list number, n number, sources list number) ? list
```

4. Exemples d'utilisation

```
import "std/c/graphbfs"  
use graphbfs  
  
create result = bfsDistances(/* args */)   
say result  
  
create result = bfsOrder(/* args */)   
say result  
  
create result = bfsReachable(/* args */)   
say result  
  
create result = bfsLayers(/* args */)   
say result  
  
create result = isConnected(/* args */)   
say result  
  
create result = shortestPathLen(/* args */)   
say result
```

5. Bonnes pratiques

- ? Préférez ``import "std/c/graphbfs"`` en tête de fichier.
- ? Lancez ``afrilang check`` avant de livrer.
- ? Combinez avec les modules `stdlib` de la même catégorie.
- ? Voir `/stdlib/` pour le source live et les démos playground.
- ? Signalez les problèmes sur GitHub si une export manque.

6. Annexe ? source

```
module graphbfs
```

```
function bfsDistances(adj list number, n number, start number) returns list number
end

function bfsOrder(adj list number, n number, start number) returns list number
end

function bfsReachable(adj list number, n number, start number) returns number
end

function bfsLayers(adj list number, n number, start number) returns list number
end

function isConnected(adj list number, n number) returns bool
end

function shortestPathLen(adj list number, n number, start number, goal number) returns number
end

function multiSourceBfs(adj list number, n number, sources list number) returns list number
end
```

ENGLISH

1. Overview

AFRILANG library `std/c/graphbfs` (tier complex, category complex). Exports 7 function(s):
bfsDistances, bfsOrder, bfsRe

```
`import "std/c/graphbfs"` . `use graphbfs`  
- `bfsDistances(adj list number, n number, start number) ? list number`  
- `bfsOrder(adj list number, n number, start number) ? list number`  
- `bfsReachable(adj list number, n number, start number) ? number`  
- `bfsLayers(adj list number, n number, start number) ? list number`  
- `isConnected(adj list number, n number) ? bool`  
- `shortestPathLen(adj list number, n number, start number, goal number) ? number`  
- `multiSourceBfs(adj list number, n number, sources list number) ? list number`
```

2. Installation & import

Add the module to your program:

```
import "std/c/graphbfs"  
use graphbfs
```

Tier: complex · Category: Complex modules (c/)
Advanced domains ? graphs, NLP, complex networks.

3. API reference

```
? bfsDistances(adj list number, n number, start number) ? list  
? bfsOrder(adj list number, n number, start number) ? list  
? bfsReachable(adj list number, n number, start number) ? number  
? bfsLayers(adj list number, n number, start number) ? list  
? isConnected(adj list number, n number) ? bool  
? shortestPathLen(adj list number, n number, start number, goal number) ? number  
? multiSourceBfs(adj list number, n number, sources list number) ? list
```

4. Usage examples

```
import "std/c/graphbfs"  
use graphbfs  
  
create result = bfsDistances(/* args */)   
say result  
  
create result = bfsOrder(/* args */)   
say result  
  
create result = bfsReachable(/* args */)   
say result  
  
create result = bfsLayers(/* args */)   
say result  
  
create result = isConnected(/* args */)   
say result  
  
create result = shortestPathLen(/* args */)   
say result
```

5. Best practices

? Prefer `import "std/c/graphbfs"` at the top of your file.
? Run `afriLang check` before shipping.
? Combine with related stdlib modules from the same category.
? See /stdlib/ for the live source and playground demos.
? Report issues on GitHub if an export is missing or undocumented.

6. Appendix ? source

```
module graphbfs
```

```
function bfsDistances(adj list number, n number, start number) returns list number
end

function bfsOrder(adj list number, n number, start number) returns list number
end

function bfsReachable(adj list number, n number, start number) returns number
end

function bfsLayers(adj list number, n number, start number) returns list number
end

function isConnected(adj list number, n number) returns bool
end

function shortestPathLen(adj list number, n number, start number, goal number) returns number
end

function multiSourceBfs(adj list number, n number, sources list number) returns list number
end
```